ENERGY OBSERVABILITY ACROSS FOURIER–GALERKIN RESOLUTIONS OF THREE-DIMENSIONAL NAVIER–STOKES

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Consider the unforced periodic three-dimensional incompressible Navier–Stokes equation after finite cubic Fourier–Galerkin truncation. This note separates two questions that are often conflated: how many simultaneous energy measurements are retained, and how deeply their time evolution must be observed before they become locally complete modulo symmetry. For every finite cubic resolution we derive sharp structural ceilings for one scalar total-energy reader and for the full shell-energy reader, and show that the resulting jet ranks are independent of every positive viscosity. Exact modular certificates attain the ceilings at the earliest structurally possible orders in four tested cases: scalar energy at $N = 1, 2$ and shell energy at $N = 1, 3$. The finite results motivate a resolution-indexed saturation conjecture and an explicit measurement–time trade-off. No continuum regularity claim is made.

1. Resolution, readers, and all-N ceilings

For $N \geq 1$, retain

$$K_N = \{-N, \ldots, N\}^3 \setminus \{0\}, \tag{1}$$

with real phase-space dimension

$$\boxed{d_N = 2\big((2N+1)^3 - 1\big).} \tag{2}$$

Write

$$\dot{x} = F_\nu(x) = \nu A x + B(x, x), \qquad \nu > 0, \tag{3}$$

where B is the projected quadratic Navier–Stokes interaction. Let

$$Q_N = \{|k|^2 : k \in K_N\}, \qquad m_N = |Q_N|, \tag{4}$$

and let $I = (I_q)_{q \in Q_N} \in \mathbb{R}^{m_N}$ denote the exact shell energies. Their sum is total retained kinetic energy

$$E = \mathbf{1}^T I. \tag{5}$$

Define the scalar and shell Lie jets

$$\mathcal{E}_R = (E, \mathcal{L}_F E, \ldots, \mathcal{L}_F^R E), \tag{6}$$

$$\mathcal{J}_R = (I, \mathcal{L}_F I, \ldots, \mathcal{L}_F^R I). \tag{7}$$

Translation by $a \in \mathbb{T}^3$ acts as $(T_a \widehat{u})_k = e^{ik\cdot a} \widehat{u}_k$. The dynamics is translation equivariant and both readers are invariant. At generic states the translation orbit is three-dimensional, so both jet ranks are at most $d_N - 3$.

The scalar jet has $R + 1$ coordinates. For the shell reader, the exact shell balance is

$$J_1 + L_\nu J_0 = \eta^{NS}, \qquad \mathbf{1}^T \eta^{NS} = 0, \tag{8}$$

where L_ν is the fixed shellwise viscous operator. Repeated differentiation yields one scalar dependence in each new m_N-component shell block. Hence:

Theorem 1 (Universal finite-resolution ceilings)**.** *For every $N \geq 1$, every $\nu > 0$, and every $R \geq 0$,*

$$\boxed{\operatorname{rank} D\mathcal{E}_R \leq \min(R+1, d_N - 3),} \tag{9}$$

$$\boxed{\operatorname{rank} D\mathcal{J}_R \leq \min\bigl(m_N + (m_N - 1)R, d_N - 3\bigr).} \tag{10}$$

Therefore the symmetry ceiling cannot be reached before

$$\boxed{R_E^{\min} = d_N - 4,} \tag{11}$$

$$\boxed{R_I^{\min} = \left\lceil \frac{d_N - 3 - m_N}{m_N - 1} \right\rceil.} \tag{12}$$

2. Positive-viscosity universality

The viscosity parameter does not generate a new rank problem at each positive value.

Theorem 2 (Positive-viscosity rank universality)**.** *Fix N and either energy reader $h \in \{E, I_q : q \in Q_N\}$. For every $r \geq 0$ and $\nu > 0$,*

$$\mathcal{L}_{F_\nu}^r h(\nu y) = \nu^{r+2} \mathcal{L}_{F_1}^r h(y), \tag{13}$$

$$D_x \mathcal{L}_{F_\nu}^r h(\nu y) = \nu^{r+1} D_y \mathcal{L}_{F_1}^r h(y). \tag{14}$$

Consequently every scalar- and shell-energy jet rank is independent of the value of $\nu > 0$.

Proof. Set $x(t) = \nu y(s)$ and $s = \nu t$. Since B is quadratic, $F_\nu(\nu y) = \nu^2 F_1(y)$. Each energy reader is quadratic, giving the displayed identities. For $\nu > 0$ the induced state and row scalings are invertible, so rank is unchanged. □

The endpoint $\nu = 0$ is different. In particular total kinetic energy is conserved for the finite Euler system, so the scalar energy jet collapses to $(E, 0, 0, \ldots)$.

3. Exact saturation certificates

The certificates propagate formal Taylor coefficients of the Galerkin flow and tangent system. For polynomial F and reader h,

$$h(\Phi_t x) = \sum_{r \geq 0} \frac{t^r}{r!} \mathcal{L}_F^r h(x). \tag{15}$$

Thus over a prime field $\mathbb{F}_p$ with $p > R$, Taylor-block and Lie-jet Jacobians have identical rank through order R. A nonzero maximal minor modulo a good prime implies that the corresponding characteristic-zero rational minor polynomial is not identically zero; the universal ceilings then give equality of the generic characteristic-zero rank.

The exact certificates currently give:

(16)

N	d_N	reader	m	$R_{\min}$	certified rank
1	52	total energy	1	48	49
1	52	shell energies	3	23	49
2	248	total energy	1	244	245
3	684	shell energies	18	39	681

Every row reaches the translation ceiling $d_N - 3$ at the earliest order allowed by the corresponding structural bound.

For $N = 2$, an exact projected Jacobian over $\mathbb{F}_{251}$ satisfies

$$\boxed{\operatorname{rank}_{\mathbb{F}_{251}} D\mathcal{E}_{244} = 245.} \tag{17}$$

For $N = 3$, the shell set is

$$Q_3 = \{1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, 14, 17, 18, 19, 22, 27\}, \tag{18}$$

so $m_3 = 18$, and an exact projected calculation over $\mathbb{F}_{683}$ gives

$$\boxed{\operatorname{rank}_{\mathbb{F}_{683}} D\mathcal{J}_{39} = 681.} \tag{19}$$

The recorded $N = 3$ milestones are

(20)

R	0	5	10	20	30	35	38	39
rank	18	103	188	358	528	613	664	681,

which follow $\min(18 + 17R, 681)$ at every recorded order.

Proposition 1 (Certified finite saturation)**.** *At the four reader–resolution pairs in the table, and for every $\nu > 0$, the corresponding energy jet has generic local rank $d_N - 3$ at its earliest structurally possible order.*

At generic states the translation action is free; for example, nonzero coefficients at the three axial modes force a fixing translation to satisfy $e^{ia_1} = e^{ia_2} = e^{ia_3} = 1$. Therefore rank $d_N - 3$ means local completeness modulo continuous translation: the kernel of the saturated jet differential equals the tangent space of the translation orbit. This is local, not global, injectivity.

4. Resolution-indexed conjecture and measurement–time trade-off

The exact cases suggest that the structural ceilings may be generically sharp at every finite resolution.

Conjecture 1 (Optimal energy-readout saturation)**.** *For every finite cubic truncation K_N and every $\nu > 0$,*

$$\boxed{\operatorname{rank}_{\text{gen}} D\mathcal{E}_{d_N-4} = d_N - 3,} \tag{21}$$

and

$$\boxed{\operatorname{rank}_{\text{gen}} D\mathcal{J}_{R_I^{\min}} = d_N - 3.} \tag{22}$$

Only the certified rows above are proved; the conjecture is not a statement about $N \to \infty$ or continuum Navier–Stokes. Nevertheless, the all-N ceilings already expose an exact design trade-off. A scalar reader requires at least

$$R_E^{\min} = d_N - 4 = 2(2N+1)^3 - 6 = \Theta(N^3) \tag{23}$$

time derivatives before symmetry-limited completeness is even possible. By contrast, the m_N-channel shell reader has lower bound (12). At $N = 3$ this changes the minimum admissible temporal depth from

$$\boxed{1 \text{ channel}: \ R \geq 680} \tag{24}$$

to the exactly saturated

$$\boxed{18 \text{ channels}: \ R = 39.} \tag{25}$$

Thus the finite theory makes precise a measurement–time principle:

$$\boxed{\text{more simultaneous dynamical readout} \ \longleftrightarrow \ \text{less temporal observability depth.}} \tag{26}$$

This does not yet identify a physical sensor layout: shell energies are spectral observables, and stable recovery from noisy high-order derivatives remains open. It does identify the quantity that a practical sensor-design program should minimize: temporal depth subject to saturating the symmetry-limited observable state.

5. Scope and verification record

The results are finite-dimensional observability statements. They do not prove continuum regularity or singularity formation, global reconstruction, stable noisy differentiation, or computational speed-up. Their precise content is the combination of universal all-N structural ceilings, positive-viscosity rank universality, and exact ceiling saturation at the certified reader–resolution pairs above.

The derivation record, executable checkers, and machine-readable certificates are maintained at

`https://github.com/morrocwi/readout-problem-navier-stokes`.

The principal records are `reproduction/NS_POSITIVE_VISCOSITY_OBSERVABILITY.md`, `reproduction/checks/check_k2_energy_observability.py`, and `reproduction/checks/check_k3_shell_energy_observability.py`.